Driver Software Requirements Analysis

Product Name	LD-1CH-PIC16F18856-T-V1.0
Document Version	V1.0
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Version No.	Reviser	Revision Date	Revision Description
V1.0	Wang Tiannan	2025/11/20	1. Define project background 2. Define resource allocation 3. Define software architecture 4. Define protection requirements 5. Define RS485 communication requirements 6. Define NFC read/write rules 7. Define dimming control

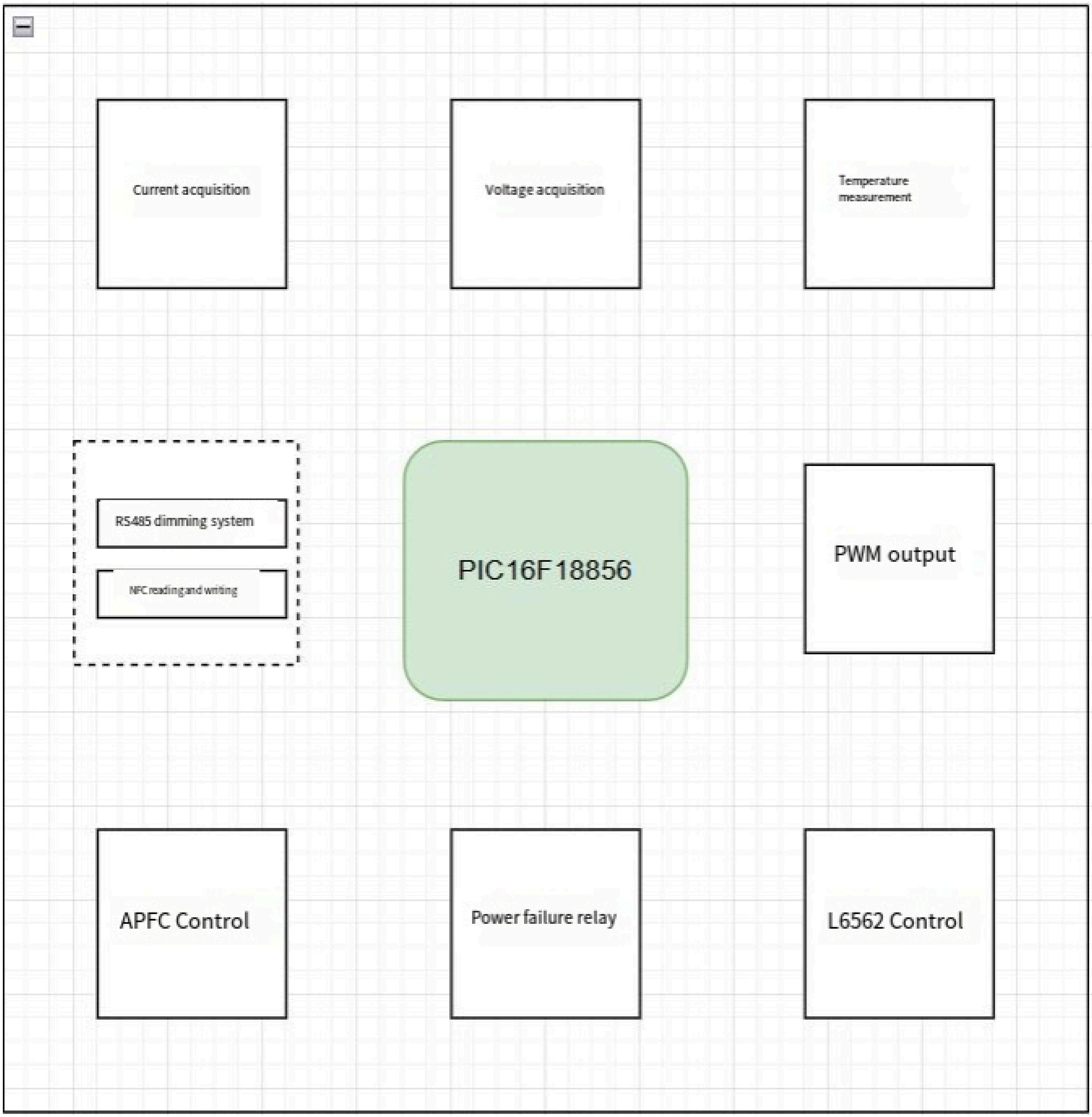
1. Overview

1.1 Project Background

This LED constant-current driver is used for agricultural lighting equipment and requires high reliability, remote communication/configuration, and multiple protection functions. The system uses an MCU to control constant current output, supports RS485 communication with a host, and supports NFC parameter configuration and command calibration.

2. System Functions

2.1 System Framework



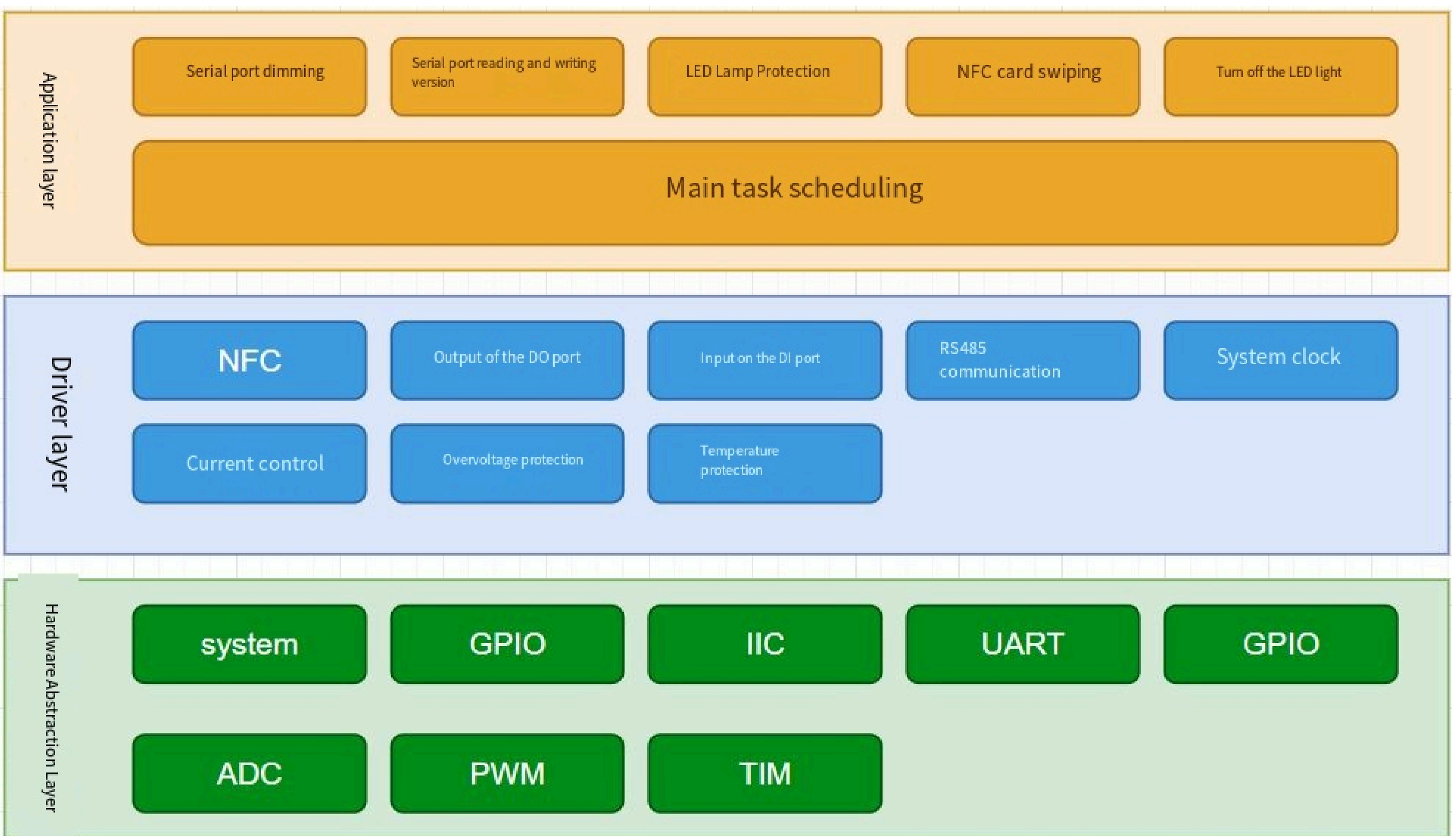
3. Hardware Resource Allocation

MCU Resources	Purpose	Notes
MCU	PIC16F18856	Main controller

MCU Resources	Purpose	Notes
ADC Sampling	Voltage/Current/Temperature sampling	10-bit resolution
PWM	Control LED output current	PWM digital output
UART	RS485 serial communication	9600 bps
NFC Read/Write	Configure basic parameters	I2C interface
GPIO	Protection detection / control outputs	Multiple outputs/inputs

4. Software Design

4.1 Software Architecture Design



4.2 Main Software Flow

1. System power-on initialization
2. Read current value and version information stored in NFC
3. Initialize RS485 communication and wait for data
4. Loop and execute each task module:

- Voltage, current, temperature sampling
- Input over/under-voltage protection
- Output short-circuit, open-circuit, over-voltage, under-voltage protection
- Temperature protection
- RS485 communication parsing
- Dimming output control

5. Functional Modules

5.1 NFC Read/Write Storage Design

Requirement Name	NFC Read/Write
Requirement Description	1. Initialize the NFC module 2. Read data and length from the specified NFC address 3. Write data and length to the specified NFC address Note: After NFC power-up it can store current values, version information, etc., which can be read/written by software.

5.2 RS485 Parsing Design

Dimming Commands

Requirement Name	Dimming Commands
Requirement Description	According to the rs_485_private_protocol.PDF command set description On serial interrupt, store each received byte into a buffer (double buffer to prevent data conflict) Parse RS485 frame header + length + data + checksum Parse the last two bytes CRCL[16], CRCH[17] and validate Read dimming ratio value to control lighting

Read/Write Version Information

Requirement Name	Read/Write Version Information
Requirement Description	On serial interrupt, store each received byte into a buffer Parse RS485 frame header[0] + address[1] Parse the last two bytes CRCL[16], CRCH[17] and validate Check function code[3] Check read/write flag[2] Logic and data processing [...] Clear buffer

5.3 Voltage and Temperature Protection Design

Input Protection

Requirement Name	Input Protection
Requirement Description	1. If input voltage > 260 VAC, allow lamp on once. 2. If input voltage > 280 VAC, normal lamp on. 3. Input undervoltage protection stage 1 (VIN < 295 VAC) reduce power. 4. Input undervoltage protection stage 2 (VIN < 280 VAC) after 3 seconds shut down PWM, disable L6562, open relay. 5. Input undervoltage recovery (VIN > 310 VAC)

Output Protection

Requirement Name	Output Protection
Requirement Description	1. Output undervoltage protection (VOUT < 150 VDC): after 1s shut down PWM, disable L6562, open relay 2. Output overvoltage protection: after 1s shut down PWM, disable L6562, open relay 3. Output short-circuit protection (after 400 ms) 4. Output open-circuit protection (after 5 s) 5. Over-power protection (power > 802 W)

Temperature Protection

Requirement Name	Temperature Protection
Requirement Description	1. Over-temperature protection stage 1 (TC > 75°C) reduce power (recover when TC < 67.5°C) 2. Over-temperature protection stage 2 (TC > 85°C) shut down PWM, disable L6562, open relay, reduce power (recover when TC < 76.5°C) 3. Temperature recovery (TC < 67.5°C) 4. 12-bit ADC, median averaging

5.4 Dimming Control

Requirement Name	PWM Dimming Control
Requirement Description	1. Control current via PWM output (0-1023 resolution) 2. Support dimming range 0x14-0xFF 3. Smooth dimming transition: fast step = 5, fine step = 1 4. Support RS485 remote dimming 5. Support NFC local dimming 2. Current values obtained from NFC per LED_NFC configuration document 3. Calibration via RS485 according to rs_485_private_protocol.PDF command specification

6. References

1. RS485 Private Communication Protocol Specification.pdf
2. LED_NFC Configuration Specification.pdf
3. PIC16F18856 Data Sheet